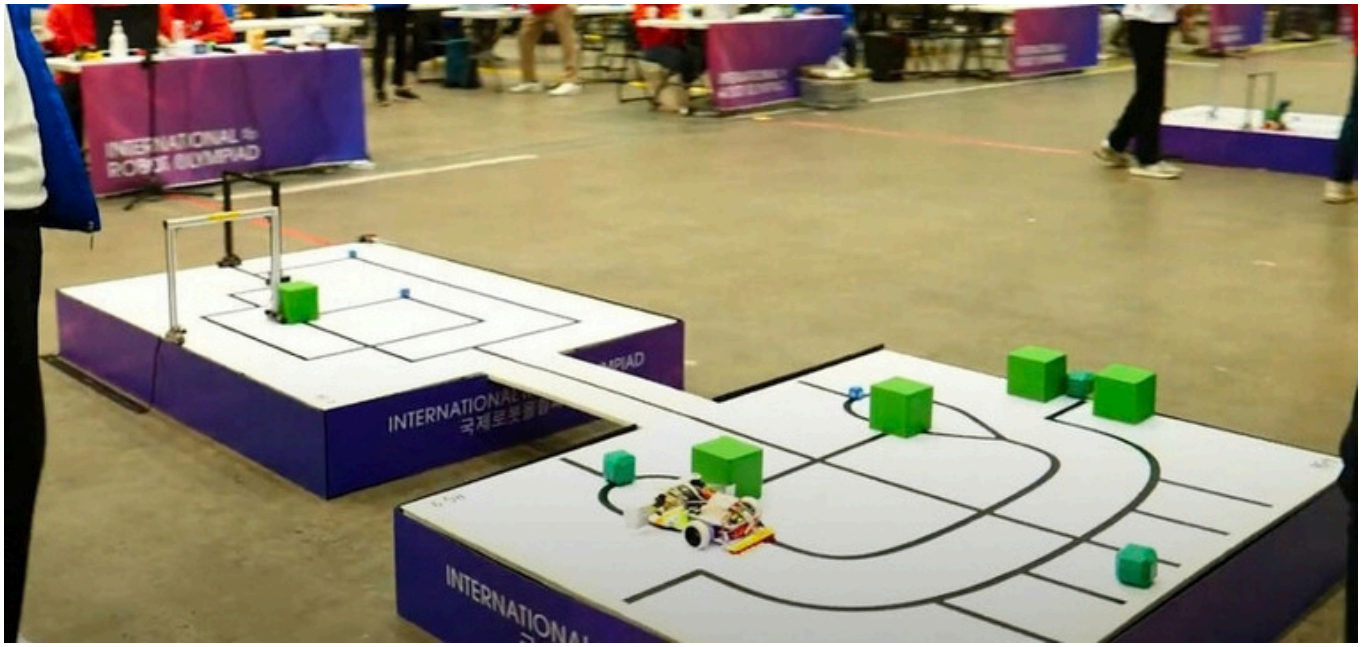
Society of Robotics and Automation, VJTI presents

SAC - SRA Autosim Challenge



Theme:

As we delve into the realm of autonomous transportation, the focus shifts towards developing advanced mechanisms that transcend the limitations of traditional line-following vehicles. While these vehicles are adept at navigating predefined paths, their applications become constrained in dynamic environments. To overcome this, engineers are exploring a spectrum of sensor technologies, including lidar, ultrasonic, and cameras, to enhance the perception capabilities of autonomous vehicles. These sensors enable vehicles to dynamically adapt to their surroundings, paving the way for more versatile and adaptable autonomous transportation systems. The increasing demand for such systems aligns with the broader trend of creating safer, more efficient, and environmentally friendly modes of transportation.

Competition Timeline:

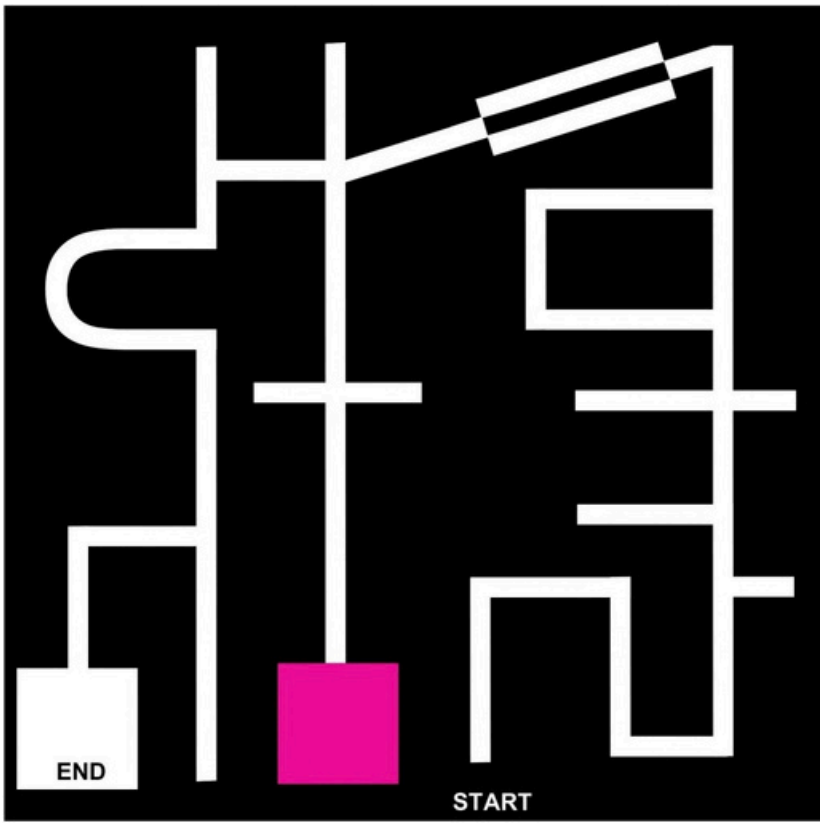
Start Date :- 07/10/2026

End Date :- 22/10/2026

Problem Statement:

SRA introduces its exclusive maze-solving competition tailored for our second-year students, presenting them with a distinctive maze to navigate and conquer.
The maze encompasses various challenges, including colorblind paths, object detection, and intricate obstacles that require strategic solutions.
Each challenge within the maze demands a logical approach for successful resolution.
The robot is tasked with commencing its journey from a designated starting point, navigating through diverse junctions, and tackling twists and turns strategically tuned to the bot's capabilities. To successfully accomplish the given tasks, participants are encouraged to leverage the available sensors on the hardware platform.

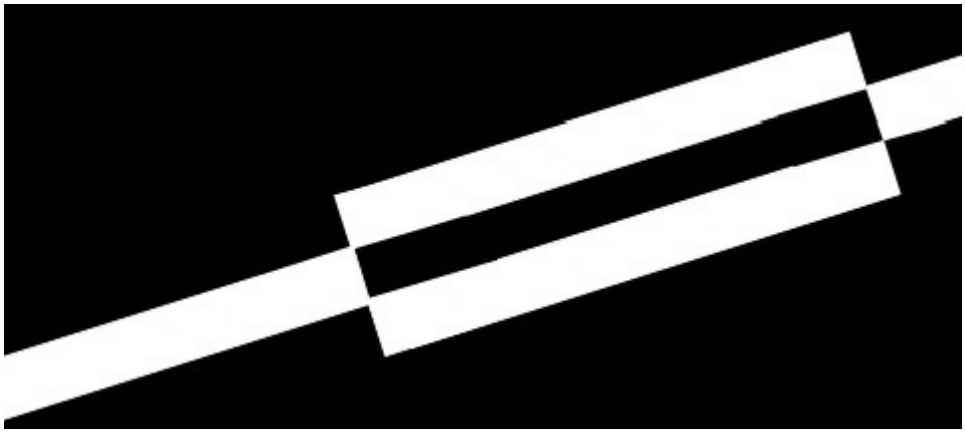
PRACTICE ARENA



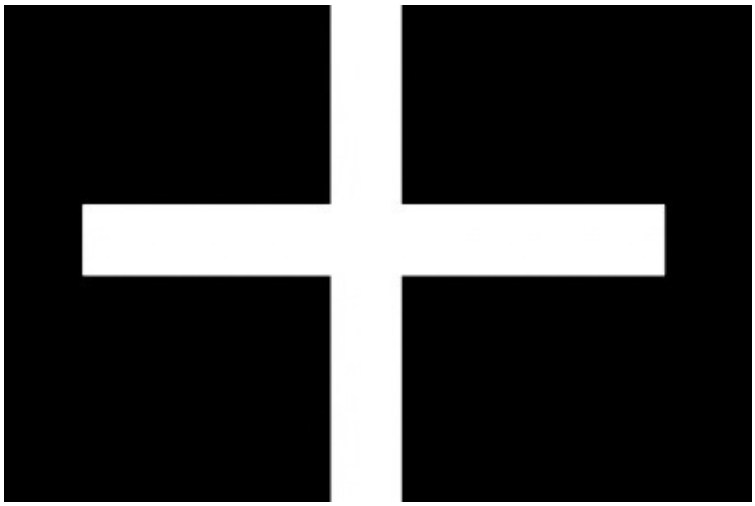
Nodes:
Then lifeline of any maze traversal are the nodes. Detect junctions, apply algorithms and keep moving forward.



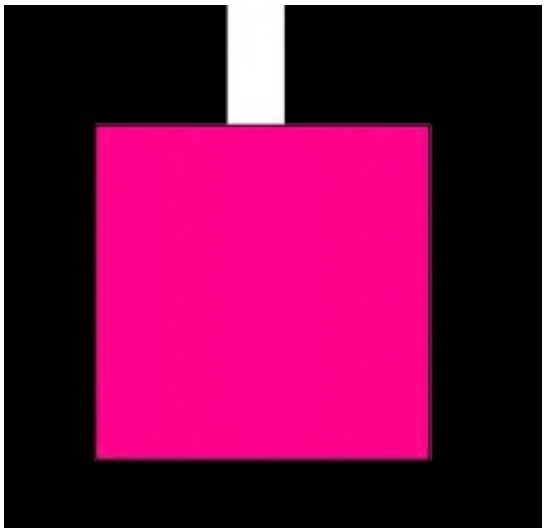
Colour Blind Path:
Does your robot work in drastically changing scenarios? This section tests your bot's adaptability to the line. Can it still stay on track if the colors are inverted?



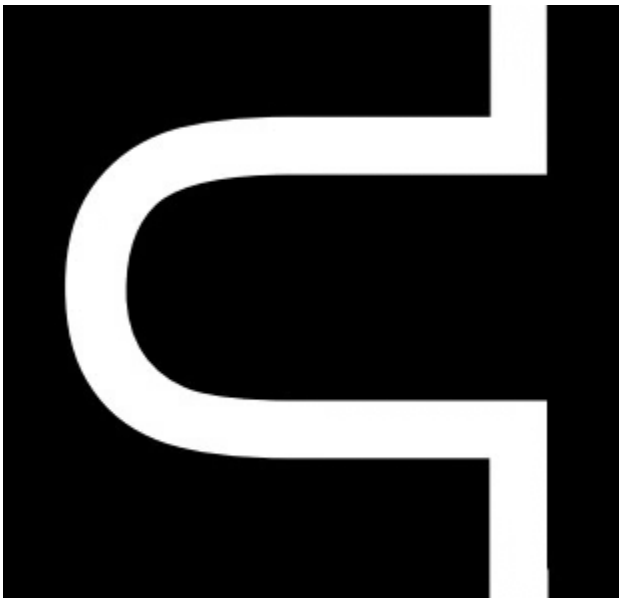
Checkpoints:
Did you go off-track? Don’t worry about having to start from scratch. If you’ve reached a checkpoint already, you can reposition your robot to the latest checkpoint. Careful though! Too many repositions and you risk being disqualified.



Object Detection Task:
Sure,your bot can follow lines, but is it useful? Can it perform real-world tasks? Detect an obstacle and push it into a Drop Zone in this crucial part of the maze.



Snake Road:
The real world isn't as forgiving as 90° turns and straight forward nodes. As the name suggests, make your bot follow a sharply curved, snake-like path using the algorithm of your choice.



GAMEPLAY:
Okso, what do you have to do actually?

1. Begin at the start zone.
2. Always adhere to the path, identifying nodes at intersection, and maintain forward movement.
3. Pass through checkpoints for potential repositioning assistance.
4. Locate the color-inverted line, follow it, and exit the zone.
5. Discover the obstacles; push them into the drop zone. Afterward, execute a U-turn and resume the maze.
6. Negotiate the sharp turn at the snake road.
7. Attain the END ZONE and bring the robot to a halt.

Repositions: If your bot drifts away from the line and continues to move in a loop/moves out of the maze, you can place the bot to the previously crossed checkpoint.
Maximum Number of re-positions allowed: TBD

Assistance: If your bot fails to follow the line (offshoots)/stops moving, you can give encouragement to the bot to bring it back to its desired position/moving state.
Maximum Number of assistance allowed: **TBD**

Scoring:

Get the maximum points by completing the maximum number of challenges in the minimum time possible.

Sr.No	Task TBD	Task code	Points
1	Checkpoint 1	C1	40
2	Checkpoint 2	C2	60
3	Checkpoint 3	C3	30
4	Checkpoint 4	C4	20
5	Pushing the obstacle * no. p	po	75*no
6	Stopping at the end zone	Stop	100

Penalties:

The first two repositions and assists are free.

1) For each extra reposition after 2 repositions : -50
2) For each extra assistance after 2 assists: -25
3) For each obstacle undetected and unpushed: -50

Example score:

Final Score = 700 + C1 + C2 + C3 + C4 + po + stop - time_taken - Penalties Hence assuming time taken to be 300 seconds in sample arena, penalty= 100: Final Score = 700 + 40 + 60 + 30 + 20 + 75*2 + 20 -300 -100 = 620

Invalid run:

1) If Max time limit is exceeded.
2) If the number of repositions is exceeded.
3) If no obstacle is detected and pushed

Tie Breaker:

In case of equal points scored by any two teams, the following points will be considered:

1. Tuning of the bot during line following
2. Time taken to complete the run
3. Code quality

General Rules:

1. The time measured by the organizers will be final and will be used for scoring the teams
2. Incase any teams are found participating in unfair practices, such as code sharing/debugging of other teams, all the involved teams will be disqualified
3. In case of any disputes/discrepancies, the organizers decision will be final and binding
4. The organizers reserve the rights to change any or all of the above rules they deem fit.
5. Teams are allowed to modify the bot with 3d printed parts or electronic components available in SRA, with sole exception of the SRA board.